



Dear Editors and Reviewers,

We thank you very much for giving us the opportunity to revise our submission. In the letter below, downloaded together with the article in the resubmission process, we reply to the reviewers.

We would like to express our surprise that the revised version of the manuscript was updated online as late as June 2026, whereas we submitted it in December 2025. In addition, we notice that we did not get feedback from Reviewer #1 about the revised manuscript, while Reviewer #2 essentially rephrased the same weaknesses, as if our point-by-point replies had not addressed them at all.

We would like to thank you again for allowing us on a path that we think has led to a much improved manuscript.

Best wishes,

Patrick Pintus, on behalf of all co-authors
Professor of Economics
Aix-Marseille University

Reply to the Review #1

Dear Reviewer #1,

Unless we are mistaken, we have not received your assessment of our point-by-point replies to all the weaknesses and recommendations, which were submitted together with the revised manuscript in the first step of the reviewing process.

It would be most useful to us, as well as to the general reader we presume, to have your feedback about whether we have satisfactorily addressed your main concerns and improved the revised version of the article that has been entirely rewritten.

Point by point replies to the Review #2 (that appear in blue)

Dear Reviewer #2,

We are not sure to understand the meaning of the first part of the review, which seems to be a repetition of the first report, with some variants and additions in the phrasing. Since we have replied in the first revision to all points raised then, simply copying and pasting here our replies, as they appeared in our first reply, seem redundant to us.

More generally, it seems unfortunate to us that the novelty of the design, approach (including some terminology) and results has received a broad negative assessment, despite our attempt to answer all your comments and recommendations in the first step.

We thank you again for allowing us on a path that we think has led to a much improved manuscript.

Best wishes,

A handwritten signature in black ink, consisting of a stylized 'P' with a vertical line extending downwards from its base.

Patrick Pintus, on behalf of all co-authors
Professor of Economics
Aix-Marseille University

Public Reviews:

Reviewer #2 (Public review):

Summary:

This paper attempts to examine how rare, extreme events impact decision-making in rats. The paper used an extensive behavioural study with rats to evaluate how the probability and magnitude of outcomes impact preference. The paper, however, provides limited evidence for the conclusions because the design did not allow for the isolation of the rare, extreme events in choice. There are many confounding factors, including the outcome variance and presence of less-rare, and less-extreme outcome in the same conditions.

Strengths.

- (1) The major strength of the paper is the significant volume of behavioural data with a reasonable sample size of 20 rats.
- (2) The paper attempts to examine losses with rats (a notoriously tricky problem with non-human animals) by substituting time-outs as a proxy for losses. This allows for mixed gambles that have both gain and loss possible outcomes.
- (3) The paper integrates both a behavioural and a modelling approach to get at the factors that drive decision-making.
- (4) The paper takes seriously the question of what it means for an event to be rare, pushing to less frequent outcomes than usually used with non-human animals.

Weaknesses:

- (1) The primary issue with this work is that the primary experimental manipulation fails to isolate the rare, extreme events in choice. As I understand the task, in all the conditions with a rare extreme event (e.g., 80 pellets with probability epsilon), there is also a less-rare, less-extreme event (e.g., 12 pellets with probability 5). In addition, the variance differs between the two conditions. So, any impact attributable to the rare, extreme event could be due to the less rare event or due difference in the variance (or other statistical moments, like skew or kurtosis). That the distributions can be shown to be different under specific assumption to value maximizing agents (e.g., with Jensen Gaps and Table 2) is not really relevant to what rats are sensitive and what drive their behaviour. The design here does not support the conclusions. Finally, by deliberately confounding rarity and extremity, the design does not allow for assessing the impact of either aspect on rat behaviour.

Reply: This is a repetition almost word for word of point (1) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(2) The RL modelling work also fails to show a specific impact of the rare extreme event. As best as I can understand Eq 2, the model provides a free parameter that adds a bonus to the value of either the two options with high-variance gains (A and V in the paper) or to the two options with high-variance losses (F and V in the paper). Or equivalently to the ones with "Jackpots" vs the ones with "Black Swans" (see Point 1 above as to how these different aspects are all confounded in this design). This parameter seems to only depends on whether this option could have possibly yielded the rare, extreme outcome (i.e., based on the generative probability) and was not connected to its actual appearance. [This point is unclear as the text says this, but the rebuttal states otherwise; plus some options never received the REE, see Table S11]. That makes it a free parameter that just bumps up (or down) the probability of selecting a pair of options. That may be due to presence of the REE or the other rare event or just the variance difference. Moreover, in the case of the "black swan" or high-variance loss conditions, this seems very much like a loss aversion parameter, but an additive one instead of a multiplicative one. Is there a theoretical claim here that "extreme losses" need an additive loss-aversion parameter?

Reply: This is a repetition almost word for word of point (2) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(3) The paper presented the methods and results with lots of neologisms and fairly obscure jargon (e.g., fragility, total REE sensitivity). That might it very hard to decipher exactly what was done and what was found. For example, on p. 4, the use of concave and convex was very hard to decipher; the text even has to repeat itself 3 times (i.e., "to repeat" and "in other words") and is still not clear. It would be much clearer (and probably accurate) to say that the options varied along the variance dimension, separately for gains and losses. Option A was low-variance gains and losses. Option B was low-variance losses and high-variance gains. Option C was high-variance losses and low-variance gains, and Option D was high-variance losses gains. That tells much more clearly what the animals experienced without the reader having to master a set of new terminologies around fragility and robustness, which brings a set of theoretical assumption unnecessarily into the description of the experimental design. Alternatively, if the authors are wary of using the term "variance" because other moments of the distribution also differ, they could use "high-value gains" or "high-value losses" or something else which does not obscure the experimental design with jargon. Again, this goes back to point 1 above, whereby the different options differ on so many dimensions (as is made even more apparent in the rebuttal) that the design cannot isolate the impact of the variables of interest.

Reply: This is a repetition almost word for word of point (3) appearing in your previous report, which

we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space. The new words that we use are an attempt at reflecting the novelty of the approach and design.

(4) Were the probabilities shuffled or truly random (seem to be fixed sequences, so neither)? What were the experienced probabilities? Given the fixed sequences, these experienced ("ex-post") probabilities, could differ tremendously from the scheduled ("ex ante") probabilities. It's quite possible that an animal never experienced the rare, extreme event for a specific option. From Table S11, that is guaranteed to have happened in that 4 animals only ever experienced the "black swan" outcome once. It's even possible (if they only picked a specific option on the 10th/60th choices by chance), that they only ever experienced that rare extreme event. This point still cannot be known given the information provided, which does not break down outcomes by options. The Supplemental in Table S11 only gives overall numbers but does not indicate what the rats experienced for each choice/option-which is what matters here. A simple table that indicates for each of the 4 options, how often they were selected, and how often the animals experienced each of the 6-8 possible outcome would make it much clearer how closely the experience matched the planned outcomes. In addition, by restricting the rare outcome to either the 10th or 60th activations in a session, these are not random. Did the animals learn this association? The text states that they did not, but no evidence is provided.

Reply: This is a repetition almost word for word of point (4) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(5) The choice data are generally presented in an overprocessed fashion with a sum and a difference (in both figures and tables). The basic datum (probability/frequency of selecting each of the 4 options) is not provided directly in the main text, even if it can theoretically be inferred from the sum and the difference. New right side of Table S4 is probably the most valuable piece in terms of explaining what rats did and should be highlighted a lot more. Inspection of that table reveals some interesting (and potentially worrying) results. Most notably, the vast majority of responding happens on the "anti-fragile" and "robust" option, often totalling around 90% of all selections, especially amongst the most common blue rats. Alas, those were all those the two options that were deliberately assigned to the two most preferred holes in the training phase (see p. 26). Does this reflect genuine preference for reward distributions or does this reflect a spatial hole bias? The assignment strategy makes this impossible to tell apart.

Reply: This is a repetition almost word for word of point (5) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space. In addition, in Figure 8 of Material and Methods, we report the changes in behavior between the training and final sessions. The related discussion stresses that half of the rats changed more than 30% of their choices in the final experimental sessions compared to the training sessions. See also our previous reply to point (8).

(6) There is insufficient detail provided on the inferential statistical tests (e.g., no degrees of freedom or effect sizes), and only limited information on exactly what tests were run and how (bootstrapping, but little detail). Without code or data (only summary information is provided in the supplement), this is difficult to evaluate. In addition, the studies seem not to be pre-registered in any way, leaving many research degrees of freedom. Not all studies need to be pre-registered and sometimes discovery of new things requires exploratory work, but preregistration does provide additional safeguards against overemphasizing post-hoc detected patterns—a serious issue in behavioural science. Moreover, this promotes transparency in reporting results and analyses, allowing for a better assessment of the strength of evidence for a claim. For example, here, were any alternative analysis pipelines attempted? Also, there were many sub-groupings of the animals and subsequent comparisons between them which all seemed post-hoc. On what grounds were these divisions made—were other divisions examined as well?

Reply: This is a repetition almost word for word of point (6) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(7) On p. 12 (Fig 4), there is an attempt to look at the impact of a rare, extreme event by plotting a measure of preference for the 10 trials before/after the rare, extreme event. In the human literature, the main impact of experiencing a rare, extreme event is what is known as the wavy recency effect (See Plonsky et al. 2015 in Psych Review for example, now cited). What this means is that there tends to be some immediate negative recency (e.g., avoiding a rare gain) followed by positive recency (e.g., chasing the rare gain). Typically, this refers to the specific option that yielded that outcome. First, as the other analyses do, the current analysis combines choice of the option that yielded the rare outcome with choice of other options, so that cannot directly assess the impact of the rare, extreme event on choice. Also, using a 10-trial window would thus obscure any impact of this rare, extreme event. There is mention of the very next trial, but an analysis that looks at the 10-trial time course trial-by-trial could reveal any impact that might be predicted from the human literature.

Reply: This is a repetition almost word for word of point (7) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(8) As I understood the method (p. 31), the assignment of options to physical locations was not random or counterbalanced, but deliberately biased to have one of the options in the preferred location. This would seem to create a bias towards a particular option and a bias away from the other options, which confounds the preference data in subsequent analyses. Table S4 reinforces this concern where the vast majority of response are clustered in the two most preferred options from training.

Reply: This is a repetition almost word for word of point (8) appearing in your previous report, which

we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(9) Are delays really losses? This is a big assumption. Magnitude and delay are different aspects of experience, which are not necessarily commensurable and can be manipulated independently. And, for the model, how were these delays transformed into outcomes for the model. Eq 1 skips over that. Is there an assumption of linearity? In addition, I was not wholly clear if the delays meant fewer trials in a session or if the delays merely extended the session and meant longer delays until the next choice period.

Reply: This is an exact repetition of point (9) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

Other points:

(1) I think the authors still misunderstand the concept of "hot-stove effects". The idea is that the experience of a very bad outcome can lead to avoiding the situation again (i.e., not sampling that option) and can provide the appearance of oversensitivity to that bad outcome. Here, that might be more thought as "black-swan avoidance". Imagine if, to the rat, all options are equal in value, then some initial bad luck in encountering the black swan might make the animal avoid that option, even though with enough experience, then it would have been equal in value.

Reply: This point is reminiscent of recommendation (5) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(2) I am still not convinced that the Jensen inequalities add to this paper in terms of understanding the rat behaviour. That may be more suited for a different paper about the statistical and mathematical properties of certain generative distributions, but not here given what rats actually choose and experience.

Reply: This point is reminiscent of recommendation (8) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(3) Providing the data open access is very good. The code, however, should be equally available and not just upon request. Code needs to be available for assessment during peer review and for reproducibility checks. There are substantial enough problems with reproducibility in the field that code availability should be a minimum criterion for publication (see Miske et al., 2026 in Nature for the most recent large-scale evaluation of this problem).

Reply: This point is reminiscent of recommendation (9) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(4) The paper still somewhat mischaracterizes the literature on rare events, posing it as a

series of "exceptions", rather than recognizing that a huge chunk of the literature uses rare events rarer than 10%. Also, there is even existing terminology in that literature for exactly the situation that is being created here-rare treasures (aka jackpots here) and rare disasters (aka Black Swans here).

Reply: This is reminiscent of point (10) appearing in your previous report, which we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(5) Defining the observed behaviour in terms convexity, instead of stating choices more plainly obscures what is done/found. This is especially the case here because convex and concave mean different things when applied to gains/losses in terms of whether or not that option can lead to the REE. The use of the terms obscures rather than clarifies and probably is best left for the discussion (and maybe the intro) when mapping from theoretical distributions to the experiment at hand. In the paper, even the bottom of p.5 seems to incorrectly define "Total Sensitivity" as the combined proportion of selecting convex options in either domain, which does not map how convex is defined in Fig 1B or elsewhere in the text.

Reply: This is reminiscent of point (3) appearing in your previous report, which as stated above we have addressed at length in our first-round reply. Therefore, we do not reiterate here to save space.

(6). Fig 1C is baffling. Why are probabilities drawn moving away from the origin? The standard scientific plotting convention is for numbers to grow when moving away from the origin. That would be vastly clearer. Also, the color coding is confusing. Green-red maps onto convex-concave, but that would naturally seem to indicate gains vs losses, not convex vs concave. And why are probabilities growing larger in both directions from the origin? Much more sensible to communicate the procedure would likely be a standard plot of magnitude vs probability.

Reply: As explained in the main text of the article, panel (c) in Figure 1 reveals an important and novel feature of the design: gains and losses are either convex or concave in **decreasing** probabilities, meaning that the gain/loss gets more rare, it becomes larger (if convex) or smaller (if concave) in a nonlinear fashion.

(7) Discussion: I think the main difference between the human situations discussed and this experiment is that humans have not experienced those rare "black swan" outcomes. Rather, they hear about the disasters that are possible and do not incorporate that information, as discussed in the description-experience literature already cited in this paper (though not in that context).

Reply: We thank the referee for pointing this out. However, it is not clear to us that the various trauma experienced in real life by humans do not share properties with Black Swans in our experimental design.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

Small points:

- p. 4. Options are not strategies. Picking two out of the 4 options more often, is not picking two strategies. That's picking two options (also on p. 7, but really throughout).

Reply: This has been changed in the text.

- The sums don't add up in probability of outcomes. How were they 0.9, 0.1, 0.01? What were they exactly? What exactly is epsilon?

Reply: Exact probabilities, which of course add up to one, are given in Material and Methods (page 29).

- The term "one-sided sensitivity" to refer to the selection of options include rare-extreme outcomes is also confusing. There are many other words that might work better here, including "encounter", "choice", "selection", or even "preference".

Reply: One-sided Sensitivity if a proposed measure of the choice asymmetry, as explained in the text (pages 5-6, including caption of Figure 1).

- I don't think that Figure 3 is more intuitive. It's still a summed measure, with rats sub-divided into post-hoc groups. Also, aren't the groups divided precisely in part by the selections that are being shown in this plot?

Reply: There is no claim that Figure 3 is more intuitive. As explained in Material and Methods (pages 33-34), it provides an alternative way to represent behaviors (BSA/JPS vs TSREE/OSREE).

- Fig 3 has no black or dark and light grey. There is no one asterisk in the figure. The light grey is irrelevant because all bars must sum to 1. No error bars or equivalent on the plot.

Reply: Dark and grey bars sum to one but have a different interpretation, as explained in the text (pages 10-11). We have dropped the one asterisk reference.

- p. 14. Technicality, but alpha need not be bound at 1 for a convergent equation. It only needs to be bound at 2 (real maximum value). Above 1 (but below 2) would be a type of overcompensation that would also converge at same Q value.

Reply: The assumption that alpha is positive and bounded by one is common in the literature and is important for stochastic convergence. Assuming that alpha is between one and two has undesirable implications, beyond possible non-convergence, in particular that Q values may fluctuate.

- p. 24. What was the generative frequency (not experienced frequency). Not adequately explained. Less than half a percent of all outcomes? Of all times that the relevant choice was made. This is not explained.

Reply: The sequences used to generate gains and losses are detailed in the Supplementary Material.

- p. 4. Says 41 sessions, but methods say 40 (e.g., Fig 7 and surrounding text)

Reply: This has been corrected in the text.

- My understanding of Tables S5 and S6 is that the best models did not include REE in Q sub-values. That seems to be opposite to the textual conclusion.

Reply: As stated in Table 1 (page 15), which summarizes results in Tables S5-S6, the selected models for rats 3, 11, and 12 do not integrate REE while the selected models for rats 4, 16 and 20 integrate REE through Q values and REE decision weights. For most rats, REE are integrated in the selected models through the latter only.

- What are the parameters in Table S7? The full 8-parameter model? But how are these the ones selected by BIC as per caption? Also, there are a number of anomalous values-lots of 1s and 0s, plus a +40 on the value of the losses for Rat 20. Columns are always gain then loss except first column.

Reply: we report in Table S7 all estimated parameters for each rat, but only for the model selected by the BIC criterion (see Table S5). Admissible values for alphas include 0 and 1. All estimates have been checked for identification. The first and second columns have been switched to have gain first.

- Table S11 does not sub-divide by option selected. So, we cannot tell if the rats actually experienced the outcome distribution associated with some options.

Reply: Given the large number of nose pokes generated by each rat, reporting all actual gains and losses would consume too much space. On the other hand, given that each option has only three frequent and rare values, it is expected that each will be experienced by each rat, in view of the choice frequencies reported in Table S2 (page 52).